

virtspawn

Quick Start Guide

How to build, install, run, and access your VM manager

Prerequisites: Fedora/RHEL/Ubuntu with libvirt + QEMU

Rust toolchain (rustup) + Node.js 18+

Estimated setup time: 5 minutes

<https://github.com/ssahani/virtspawn>

Step 1: Clone and Build

1

Clone the repository



```
$ git clone https://github.com/ssahani/-virtspawn.git
```

```
$ cd virtspawn
```

2

Build everything (Rust backend + React frontend)



```
# Build Rust binaries (daemon + TUI) in release mode
```

```
$ make release
```

```
# Build web frontend
```

```
$ make web
```

```
# Or build everything in one command
```

```
$ make all
```

Build output



```
target/release/virtspawn-daemon # REST API daemon (~10 MB)
```

```
target/release/virtspawn # Terminal TUI (~8 MB)
```

```
web/dist/ # Web UI static files
```

A One-command deploy (install + start)



```
$ sudo make deploy
```

Installs to `/usr/local/bin/`

Web UI to `/usr/local/share/virtspawn/web/`

Config to `/etc/virtspawn/config.toml`

Enables and starts `systemd` service

B Manual step-by-step



```
$ sudo make install # Install binaries and files
```

```
$ sudo systemctl start virtspawn-daemon # Start the daemon
```

```
$ sudo systemctl enable virtspawn-daemon # Auto-start on boot
```

Verify it is running:



```
$ sudo systemctl status virtspawn-daemon
```

Active: active (running) ... listening on 0.0.0.0:8081

Step 3: Access virtspawn

Web UI

Open in your browser:

`http://localhost8081`

Full dashboard with charts,

VM management, storage,

networks, and console access.

Terminal TUI

Run from any terminal:

`$ virtspawn`

Keyboard-driven interface.

j/k to navigate, s to start.

x to stop, ? for help.

REST API

curl from scripts/tools:

`$ curl localhost8081/api/v1/vms`

`$ curl localhost8081/api/v1/node`

30+ JSON endpoints for

full automation.

Remote access: replace localhost with your server IP

Default port 8081 | Config: /etc/virtspawn/config.toml

All three interfaces connect to the same daemon

Method	Endpoint	Description
GET	/api/v1/health	Health check
GET	/api/v1/vms	List all VMs
GET	/api/v1/vms/{name}	VM details
POST	/api/v1/vms	Create VM
POST	/api/v1/vms/{name}/start	Start VM
POST	/api/v1/vms/{name}/shutdown	Graceful shutdown
DELETE	/api/v1/vms/{name}	Delete VM
GET	/api/v1/networks	List networks
GET	/api/v1/storage/pools	List storage pools
GET	/api/v1/metrics	Live VM metrics
GET	/api/v1/node	Host node info
GET	/api/v1/capabilities	Hypervisor capabilities
POST	/api/v1/vms/{name}/migrate	Migrate VM

Full API: 30+ endpoints covering VMs, networks, storage, snapshots, devices, and more

All responses are JSON | Errors include descriptive messages

Quick Reference



Build

```
$ git clone https://github.com/ssahani/-virtspawn.git
```

```
$ cd virtspawn && make all
```

Deploy

```
$ sudo make deploy
```

Access

```
Web UI: http://localhost8081
```

```
TUI: virtspawn
```

```
API: curl http://localhost8081/api/v1/health
```

Manage

```
$ sudo make status | start | stop | restart
```

virtspawn -- Modern Libvirt VM Manager

Built with Rust + React | Secure | Fast | Production-Ready

<https://github.com/ssahani/-virtspawn>